Mars Habitat Automation System

A distributed IoT monitoring and control platform for Mars habitat simulation, built for the Advanced Programming Laboratory course.

System Architecture Overview

1

Data Ingestion

Sensor data collected from mars-simulator via REST and SSE endpoints

2

Message Routing

Normalized messages distributed via RabbitMQ message broker

3

Automation Engine

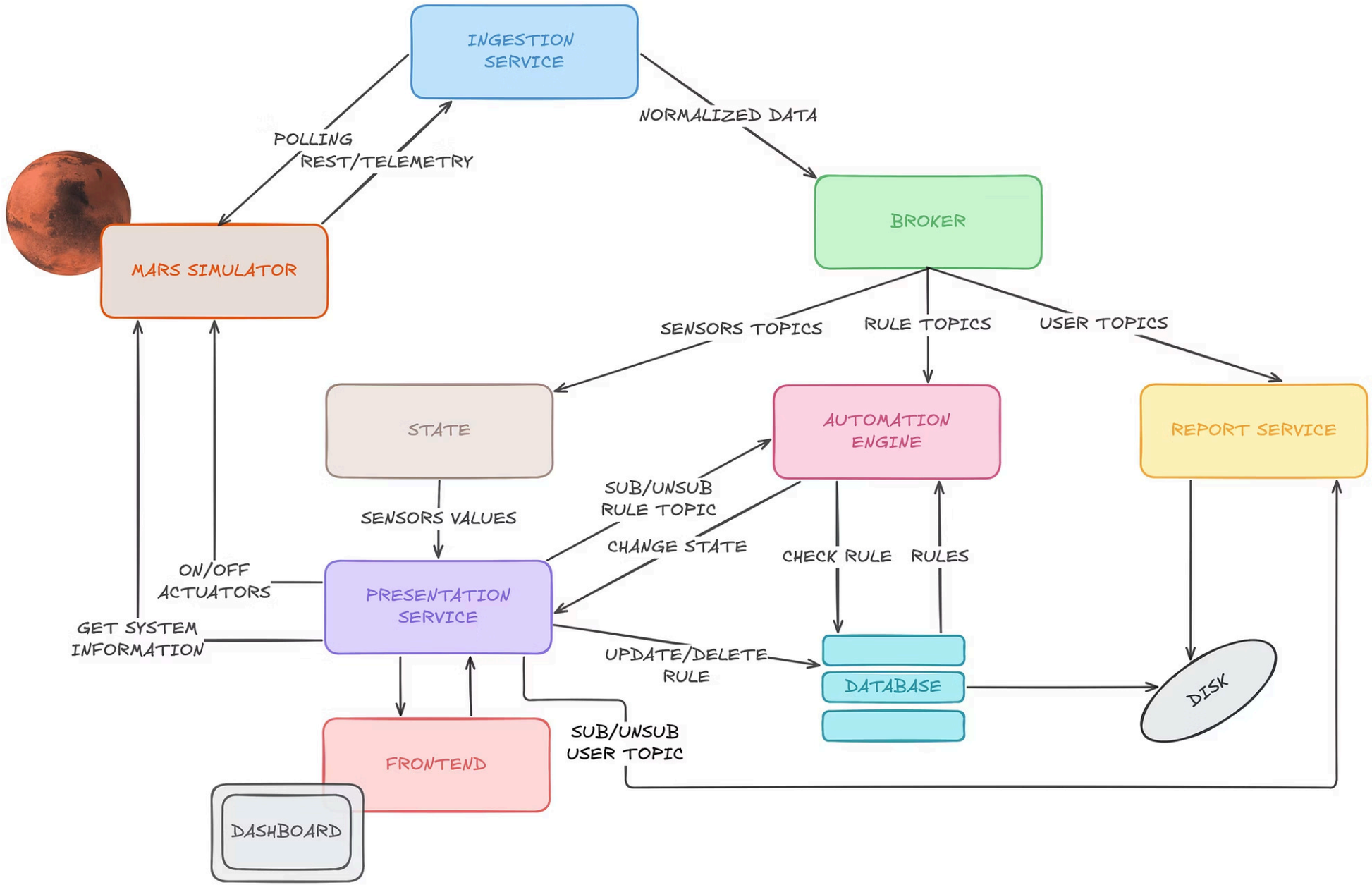
Rules applied to trigger actuator responses based on sensor conditions

4

Dashboard UI

Real-time monitoring and manual control through web interface

Architecture



Technology Stack

Message Broker

RabbitMQ with topic-based exchanges for real-time data distribution across services

Data Persistence

PostgreSQL database storing automation rules with CRUD operations for rule management

Core Services

Python 3.11 microservices handling ingestion, automation, state management, and reporting

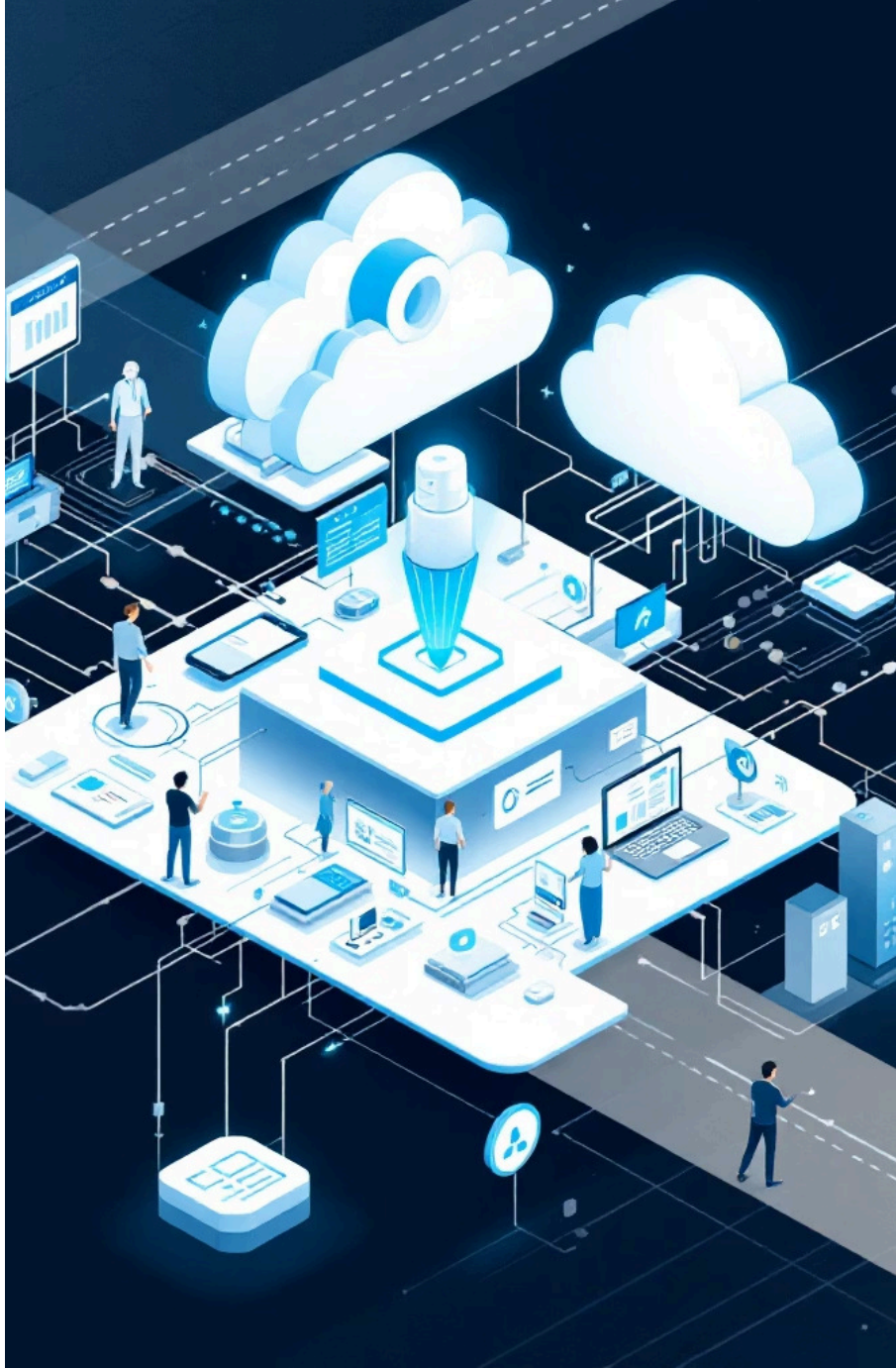
Web Interface

Flask + Flask-SocketIO backend with HTML/JS frontend for real-time dashboard and controls

Event Schema Normalisation

To handle incompatible device dialects, all incoming data is normalised into a unified internal format before being published to the message broker.

```
{
 "id": "string",
 "source": "rest | telemetry",
 "captured_at": "ISO_8601 timestamp",
 "status": "ok | warning",
 "measurements": [
 {
 "metric": "string",
 "value": "number",
 "unit": "string"
 }
 ],
 "topic": "string"
}
```



Rule Model & Automation

Dynamic Rules

IF-THEN rules evaluated upon arrival of new sensor events

Database Persistence

Rules stored in PostgreSQL to survive service restarts

Actuator Control

Automatic ON/OFF switching based on threshold conditions

Monitoring User Stories

View Sensors

Monitor available sensors with current values and measurement units

Real-Time Telemetry

View live line charts for telemetry data including power and radiation

Sensor Warnings

Visual alerts when sensors exceed critical thresholds or become unreachable

Actuator Control Stories

Manual Control

Operators can manually activate or deactivate actuators through the dashboard interface, providing override capability when automation needs to be bypassed.

Automated Responses

System automatically triggers actuators when rule conditions are met, maintaining optimal environmental conditions without manual intervention.

View actuator list and current states

Track activation history through system logs

Rule Management Stories

Create Rules

Build new automation rules that react to sensor values and control environmental conditions

Update Rules

Modify existing rules to respond to changing habitat conditions and requirements

Delete Rules

Remove outdated automation logic that is no longer applicable to current operations

Enable/Disable

Temporarily deactivate rules without deleting them for testing and troubleshooting

System Features & Capabilities

Rule Persistence

Automation logic preserved and restored automatically after system reboots or failures

PDF Reports

Generate detailed sensor charts and statistics for archiving and analysis

Action Logging

Comprehensive history of actuator state changes for auditing automation behaviour

Key Achievements

7

Microservices

Containerised services working in harmony

16

User Stories

Complete feature coverage for operators

99.9%

Real-Time

WebSocket updates for live monitoring

This distributed automation platform demonstrates event-driven architecture principles, microservices communication patterns, and real-time system monitoring capabilities suitable for mission-critical environments.